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D&A – Snowflake & IDMC

Table of Content



Week	Day	Topic & Skill	Content Coverage
1	1	Programming Fundamentals + Database & RDBMS	Planning Logic – Pseudocode, flowcharts Programming Basics – What is programming, types of languages Syntax & Structure – Variables, data types, operators Control Structures – if, else, loops (for, while) Functions and Modular Programming – Defining, calling, parameters, return values Database: Introduction to Databases What is a Database? Types of Databases: Relational vs Non-Relational Overview of RDBMS (Relational Database Management System) Popular RDBMS Tools: MySQL, PostgreSQL, Oracle, SQL Server, Teradata SQL Basics SQL Syntax and Structure Data Definition Language (DDL): CREATE, ALTER, DROP, TRUNCATE
	2	Database & RDBMS	Database contd.: Data Manipulation Language (DML): INSERT, UPDATE, DELETE Data Query Language (DQL): SELECT, WHERE, FROM, ORDER BY, GROUP BY Advanced SQL Joins: Inner, Left, Right, Full [Focus] Subqueries and Nested Queries Transactions and Concurrency ACID Properties Transaction Control: COMMIT, ROLLBACK, SAVEPOINT [HoL must] Database Administration - server backups User Roles and Permissions
	3	SQL Fundamentals & Intermediate SQL	Operators: AND, OR, NOT, BETWEEN, IN, LIKE Recap Joins: INNER, LEFT, RIGHT, FULL OUTER, SELF JOIN Recap Subqueries and Nested Queries [Focus] Aliases and Set Operations (UNION, INTERSECT, MINUS) Aggregate Functions: COUNT, SUM, AVG, MIN, MAX Views and Indexing Data Control Language (DCL): GRANT, REVOKE DQL - HAVING, LIMIT

Week	Day	Topic & Skill	Content Coverage
	4	Advanced SQL Techniques	Advanced SQL Techniques Window Functions: ROW_NUMBER(), RANK(), LEAD(), LAG() Case Expressions and Conditional Logic Stored Procedures and Triggers
	5	SQL Practical Applications	Practical Applications Real-world Projects: Inventory System/Analytics Dashboard SQL for Data Analytics and Reporting Working with Large Datasets
2	1	Advanced Analytical Topics & Recursive CTE **+ Weekly Assessment-1**	Advanced Analytical Topics Query Optimization and Execution Plans Temporary Tables and Clustering Common Table Expressions (CTEs) Recursive CTEs Recursive CTE for Organizational Hierarchies
	2	Performance Benchmarking & DWH Concepts	Recursive CTE for Organizational Hierarchies Performance Benchmarking Data Warehouse (DWH) Concepts Introduction to Data Warehousing What is a Data Warehouse? OLTP vs OLAP systems Benefits and use cases of DWH Key components of a DWH system Introduction to snowflake Data warehouse lifecycle [SQL on snowflake labs needed - single database]

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	3	DWH Architecture & data modelling	Data Warehouse Architecture Basic architecture: staging, integration, access layers Enterprise Data Warehouse (EDW) vs Data Marts ETL vs ELT processes Real-time vs batch data warehousing [batch data lab needed] Cloud-based DWH (e.g., Snowflake, BigQuery, Redshift, Databricks) Data Modeling Dimensional modeling: Star and Snowflake schemas [Lab needed] Fact and Dimension tables Slowly Changing Dimensions (SCD Types 1, 2, 3) Granularity and surrogate keys Normalization vs denormalization SQL on snowflake labs needed - multiple databases (fact/dimension tables implementation)
	4	ETL/ELT & data profiling	ETL/ELT Processes Overview of ETL tools (Informatica, Qlik Talend, Fivetran, Matillion etc.) Data extraction from source systems Data transformation techniques Data loading strategies Error handling and logging Data Integration, Quality and BI concepts Data profiling and cleansing Handling duplicates and missing data [Compute management lab needed] [Query + visualization lab needed]
	5	Cloud Data Warehousing	Cloud Data Warehousing Introduction to cloud DWH platforms Snowflake, Databricks and Google BigQuery and Redshift overview Scalability Sample Real-World Projects
3	1	**Mocktest-1 and Debriefing**	

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	2	Python Fundamentals	Introduction to Python and its ecosystem Installing Python and setting up IDEs (VS Code, PyCharm, Jupyter) Basic syntax, variables, and data types - int, str, float, var, list Operators and expressions Input/output operations Comments and code documentation
	3	Control Structures & Loops	Data Structures Lists, Tuples Control Structures Conditional statements (if, elif, else) Loops (for, while) Loop control (break, continue, pass) Comprehensions (list, dict, set) Regular Expressions
	4	Functions and Modules	Functions and Modules Defining and calling functions Arguments and return values Lambda functions Recursion Modules and packages Importing and using libraries
	5	Data Structures	Data Structures Recap - Lists, Tuples Set, Dictionaries Operations and methods Nested structures Iteration and manipulation
4	1	Object-Oriented Programming (OOP) Working with libraries	Object-Oriented Programming (OOP) Classes and objects Constructors and destructors Inheritance and polymorphism Encapsulation and abstraction Magic methods (__init__, __str__, etc.) Working with Libraries NumPy for numerical data analysis [numpy focus]
	2	Numpy & Pandas	NumPy and Pandas for data manipulation [pandas focus]

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	3	Working with libraries - Matplotlib, Seaborn, SQLite/SQLAlchemy	Matplotlib and Seaborn for visualization SQLite/SQLAlchemy for database interaction [1 SQL connectivity lab needed]
	4	**Weekly Assessment-2 +** Exception & File Handling	Exception Handling Try, Except, Finally blocks Custom exceptions Assertions Logging and debugging File Handling Reading and writing files Working with CSV, JSON, XML File modes and context managers Directory and file operations
	5	Testing & deployment Data Analysis Project	Testing and Deployment Unit testing with unit test and pytest Data analysis project (e.g., COVID trends) Automation scripts (e.g., email sender, file organizer)
5	1	Data Analysis Project	Data analysis project (e.g., COVID trends) Automation scripts (e.g., email sender, file organizer)
	2	AI Essentials	Essential AI Topics Introduction to AI - What is AI, ML, DL, AGI Prompt Engineering & ChatGPT - What is prompt engineering - ChatGPT use cases (e.g., for data analysis) Building Generative AI Applications - Prototyping and deploying apps - Evaluation techniques for NLP models
	3	**Mocktest-2 and Debriefing**	
	4	**Milestone 1 Assessment**	
	5	Snowflake Core - Introduction & Setup	1. Introduction & Setup Snowflake architecture: storage, compute, services Supported clouds: AWS, Azure, GCP Account creation, UI navigation, roles & permissions

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6	1	Snowflake Core - SQL Data Operations	2. SQL & Data Operations Snowflake SQL: DDL, DML, SELECT, joins, functions Data types, views, stored procedures
	2	Snowflake Core - Data Loading & Unloading **+ Milestone 1 Retest (If any)**	3. Data Loading & Unloading File formats: CSV, JSON, Parquet COPY INTO, Snowpipe, staging Unloading to cloud storage
	3	Snowflake Core - Performance & Optimization	4. Performance & Optimization Virtual warehouses, caching Query profiling, clustering keys
	4	Snowflake Core - Security & Sharing	5. Security & Sharing RBAC, encryption, network policies Time Travel, fail-safe Secure data sharing
	5	Snowflake Core - Semi-Structured Data	6. Semi-Structured Data JSON, Avro, Parquet, XML VARIANT type, flattening nested data
7	1	Snowflake Core - Integrations & Pipelines and Account Management	7. Integrations & Pipelines BI tools (Tableau, Power BI), ETL tools (dbt, Fivetran) Streams, tasks, materialized views Batch & streaming ingestion 8. Account Management Billing, credits, resource monitors
	2	Snowflake Advanced - Snowpark **+ Weekly Assessment 3**	9. Snowpark APIs for Python, Java, DataFrames, UDFs, stored procedures
	3	Snowflake Advanced - Snowflake Cortex	10. Snowflake Cortex AI/ML functions, Cortex Agents & Search AISQL, REST API integration Secure AI dashboards
	4	Snowflake Advanced - Streamlit in Snowflake	11. Streamlit in Snowflake Build interactive apps Visualizations with Matplotlib, Seaborn SQL + Snowpark integration
	5	Snowflake Advanced - Snowsight and Overview of Snowflake Horizon	12. Snowsight Modern UI for SQL, dashboards, Streamlit Monitoring usage, governance tools 13. Overview of Snowflake Horizon

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8	1	SnowAdvanced Architect Overview	Accounts and Security Snowflake Architecture
	2	SnowAdvanced Architect Overview	Data Engineering ELT/ETL pipeline design using Snowpipe, Streams, Tasks
	3	SnowAdvanced Architect Overview	Handling semi-structured data (JSON, Parquet) UDFs, stored procedures, dynamic tables Integration with tools like dbt, Kafka
	4	**MS 2 - Mock Assessment**	
	5	**Milestone 2 Assessment**	
9	1	IDMC - Foundation	1. IDMC Foundation Level Basics of CDI, CDQ, and CAI Mapping, profiling, process design
	2	IDMC - Data Integration & transformations	2. Data Integration & Transformations Mappings, reusable objects Joiner, Aggregator
	3	IDMC - Data Integration & transformations **+ Milestone 2 Retest (If any)**	2. Data Integration & Transformations Lookup, Router Hierarchical data handling
	4	IDMC - ELT & Scheduling	3. ETL & Scheduling Batch/real-time ingestion, CDC SQL pushdown, cloud warehouse integration Taskflows(email, command, decision and parallel), triggers, listeners Parameterization
	5	IDMC - Monitoring	4. Monitoring Logs, error handling, performance tuning
10	1	IDMC - CDI **+ Weekly Assessment 4**	5. Cloud Data Integration (CDI) Track Build and manage pipelines across hybrid/multi-cloud Mapping designer, transformations, ingestion, ELT, orchestration
	2	IDMC - CAI	6. Cloud Application Integration (CAI) Track CAI architecture, API management Process designer, service connectors Fault handling, monitoring, asset management

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	3	IDMC - CAI-CDI Integration	7. CAI-CDI Integration Invoke CDI mappings from CAI End-to-end orchestration
	4	Snowpro Core Certification and IDMC certification - Walkthrough Day	
	5	**MS 3 - Mock Assessment**	
11	1	**Milestone 3 Assessment**	
	2	**Client Interview Simulation**	
	3	Capstone Project	
	4	Capstone Project **+ Milestone 3 Retest (If any)**	
	5	Capstone Project	
12	1	Capstone Project	
	2	Capstone Project	
	3	**Milestone 5 Capstone Project Evaluation**	
	4		
	5		
13	1	**Capstone Project Re-Evaluation**	

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It's time to Outcreate

